ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. 594 documents,

2. Filtering the dataset by category helps frequent pattern mining focus on a specific topic. It reduces the influence of unrelated documents and prevents important medical terms from being overlooked when analyzing the entire dataset. As a result, the discovered patterns are more relevant and meaningful for medical-related research.

AI used for this answer: Yes

If yes, how: Concept explanation and English editing

External references (links):

https://scikit-learn.org/stable/datasets/real_world.html#the-20-newsgroups-text-dataset

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Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Dropping 40% of the documents would result in a significant loss of useful text data. It could also introduce selection bias if the missing labels are related to certain characteristics of the documents. Instead of removing these records, we can use the existing numerical category labels to recover the missing category names, assuming those labels are available and reliable.

AI used for this answer: Yes

If yes, how: Concept explanation and English editing

External references (links):

https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html?utm_source=chatgpt.com

https://pandas.pydata.org/docs/reference/api/pandas.Series.map.html?utm_source=chatgpt.com

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

The category proportions in the sample do not exactly match those in the original dataset. For example, the proportion of comp.graphics increases from 25.88% to 27.80%, while soc.religion.christian decreases from 26.54% to 24.80%.

For highly imbalanced datasets, simple random sampling may select too few documents from minority categories or even miss them entirely. This could make the sample less representative and affect the analysis results. Stratified sampling can help preserve the original category proportions.

AI used for this answer: Yes

If yes, how: Concept explanation and English editing.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.StratifiedShuffleSplit.html?utm_source=chatgpt.com

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

When changing stop_words='english' to stop_words=None, the number of features increased from 35,482 to 35,788 because common English stopwords were included.

The sparsity decreased slightly from 99.6772% to 99.5470%. This is because stopwords appear frequently across documents, adding more non-zero values to the document-term matrix and reducing the overall percentage of zeros.

AI used for this answer: Yes

If yes, how: Concept explanation and experimental result interpretation.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html?utm_source=chatgpt.com

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Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimens**ionality wi**thout harming model accuracy?

Answer:

As the variance threshold increases from 0.0 to 0.05, the number of retained features decreases significantly, from 35,482 to 1,491.

Words with near-zero variance are usually very rare words that appear in only a few documents, or words that occur with almost the same frequency across documents. These features are removed first because they show very little variation.

Removing low-variance features helps reduce dimensionality and computational costs, since these features usually provide limited information for distinguishing documents. However, it does not guarantee that model accuracy will always remain unchanged, because some rare words may still be useful for classification.

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

AI used for this answer: Yes

If yes, how: Concept explanation and English editing.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html?utm_source=chatgpt.com

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Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. Top-correlated words

The top-correlated words include both domain-specific terms and general words. Pearson identifies words such as "vga," "video," and "card," but also general words like "thanks" and "writes." Spearman identifies more graphics-related terms, such as "graphics," "image," "3d," and "program." This shows that high statistical correlation does not necessarily mean a word is domain-specific.

2. Pearson vs. Spearman

Pearson measures the linear relationship between the original word frequencies and the target category, while Spearman measures the monotonic relationship using ranks. Therefore, Spearman is generally less sensitive to extreme values than Pearson.

3. When Spearman can be higher than Pearson

For example, a graphics-related word may appear more frequently in graphics documents, but a few documents may contain extremely high word counts. These extreme values can weaken the Pearson correlation, while Spearman may still capture the overall monotonic relationship. In our results, the word "graphics" has a Pearson correlation of 0.1779 but a Spearman correlation of 0.5211. However, this difference alone does not prove that outliers are present.

AI used for this answer: Yes

If yes, how:

Concept discussion, interpretation of experimental results, English translation and lang uage editing.

External references (links):

https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html?utm_source=chatgpt.com

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup

When minSup decreases from 9 to 3, the number of discovered maximal frequent patterns increases significantly from 710 to 9,547. The runtime also increases from 0.2015 to 0.6287 seconds.

This happens because a lower minimum support threshold allows more itemsets to qualify as frequent, expanding the search space and increasing computational effort. In this experiment, both the number of discovered patterns and the execution time increased.

Answer (2) — Top-K k vs. MaxFPGrowth minSup

The parameter k in Top-K FAE specifies how many of the most frequent patterns to retrieve, ranked by support. In our experiment, setting k to 100 and 500 produced 100 and 500 patterns, respectively.

In contrast, minSup in MaxFPGrowth specifies the minimum number of transactions in which an itemset must appear. The algorithm then identifies maximal frequent patterns that satisfy this threshold. Therefore, k controls the target number of patterns, while minSup controls the minimum frequency required for a pattern to qualify.

Answer (3) — Maximal vs. standard frequent patterns

Standard frequent pattern mining can return all itemsets that satisfy the minimum support threshold, including many subsets of larger frequent itemsets.

Maximal frequent pattern mining only returns frequent itemsets that are not subsets of any larger frequent itemset. By excluding these redundant subsets, it produces a more concise representation of the frequent patterns. However, the exact support of every excluded subset cannot necessarily be recovered from maximal patterns alone.

AI used for this answer: Yes

If yes, how:

Concept discussion, interpretation of experimental results, and English drafting and editing based on my understanding.

External references (links):

https://github.com/UdayLab/PAMI?utm_source=chatgpt.com

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

Answer (1)

The original X_counts matrix contains 2,257 documents and 35,482 word features. A 2D scatter plot requires only two coordinates for each document, so we cannot directly visualize all 35,482 dimensions in a single 2D plot. Selecting just two original features would ignore most of the information in the dataset.

Answer (2)

We should first apply dimensionality reduction algorithms, such as PCA, t-SNE, or UMAP, to transform the high-dimensional text data into a 2D representation. This allows us to visualize the distribution of the four document categories while preserving useful aspects of the original data structure.

AI used for this answer: Yes

If yes, how:

Concept discussion and English drafting based on my understanding.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html?utm_source=chatgpt.com

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html?utm_source=chatgpt.com

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?

2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. Comparison of cluster separation

PCA shows heavy overlap among the four document categories, with most points concentrated near the center. In contrast, t-SNE and UMAP reveal more visible local clusters and partial separation between categories. However, the four categories are not completely separated in either visualization.

2. Why PCA struggles with high-dimensional text data

PCA is a linear dimensionality reduction method that preserves directions of maximum overall variance, which may not correspond to differences between document categories. In contrast, t-SNE and UMAP focus more on preserving local neighborhood relationships and can reveal nonlinear structures that PCA may fail to capture.

3. Limitations of interpreting t-SNE distances

t-SNE mainly preserves local similarities rather than exact global distances. Therefore, clusters that appear far apart in a 2D t-SNE plot are not necessarily equally far apart in the original high-dimensional space. Analysts should avoid interpreting the physical distances between clusters as precise measures of their original dissimilarity.

AI used for this answer: Yes

If yes, how:

Discussion of visualization results, conceptual explanation, and English drafting based o n my observations.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html?utm_source=chatgpt.com

https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html?utm_source=chatgpt.com

https://umap-learn.readthedocs.io/en/latest/?utm_source=chatgpt.com

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Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

Answer (1)
Since the dataset contains four categories, LabelBinarizer produces four binary columns. The resulting encoded matrix has a shape of (2257, 4), where each document is represented by a binary vector with exactly one value of 1.
Answer (2)
Using the original integer labels (0, 1, 2, 3) as numerical input features may incorrectly imply that the categories have a natural order or meaningful numerical distances. However, these numbers are only category identifiers, so their differences do not represent actual relationships between the categories.
Answer (3)
One-hot encoding represents each category using a separate binary position, with 1 indicating the corresponding category and 0 indicating the others. This prevents models from interpreting arbitrary category labels as ordered numerical values and avoids introducing misleading distances between categories.

AI used for this answer: Yes

If yes, how:

Concept discussion, interpretation of experimental results, and English drafting based o n my understanding.

External references (links):

https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.LabelBinarizer.html?utm_source=chatgpt.com

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